Analytical fan-beam reconstruction algorithm for free-form trajectory with plus-minus weighting scheme

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ABSTRACT

This paper focuses on fan-beam image reconstruction from free-form X-ray source trajectory in computed tomography (CT). As in major standard image reconstruction methods, the weighting function of the redundant projection data should be carefully considered. The data redundancy reduction principle aims to compute the projection data only one time during image reconstruction. Usually, an X-ray line that passes through the object point intersects with the radiation source trajectory more than once time. For the symmetric closed circular trajectory, because each X-ray line passing through an object point would intersect with the source trajectory twice, the weighting function of 0.5 is used to handle this problem. However, normally there is no known symmetry property for the free-form X-ray source trajectory. In order to estimate the weighting function we have to calculate the number *N* of intersection between each X-ray line and every object points, then the weighting function is set to be 1/N such that the summation of weighting operator corresponding to a single X-ray line is the unity. However, the calculation of *N* is difficult and computationally expensive. In this paper we proposed a new scheme for a robust control of the redundant projection data for both closed and open trajectories. Instead of calculation of N, we assign each intersection point a plus or minus sign according to the proposed weighting function. As a result of summation of successive signs, they cancel out each other and finally equal to the unity. Numerical study was performed to evaluate the proposed weighting scheme using standard Shepp-Logan phantom with butterfly trajectory.

Keywords: Computed tomography, fan-beam reconstruction, weighting scheme, free-form trajectory

1. INTRODUCTION

In most applications of two-dimensional computed tomography (2DCT), the closed circular X-Ray source trajectory is used because of its simplicity in implementation. However, the circular trajectory is not always the most appropriate choice for different reconstruction objects. For example, the elliptical trajectory is much better for rectangular object as it can lead to reduction in scanning time and radiation exposure. In some special imaging applications such as cardiac CT, because the cardiac is beating in high and irregular rates during projection data acquisition, the projection data could not be directly used for reconstruction. However, if a rigid transformation is applied on the circular trajectory, we can suppose the cardiac is motionless and reconstruct it from an irregular and free-form trajectory. In theory, several algorithms have been proposed for image reconstruction in fan-beam case in [1-5]. Based on circular trajectory the reconstruction algorithms were proposed in [1-3]. The algorithms in [4] and [5] considered in special noncircular trajectory case. A general fan-beam image reconstruction algorithm for free-form X-Ray source trajectory has a significant impact on practice application and theory. In this paper, we rewrote the fan-beam reconstruction formula presented in [1], such that it can be applied not only to circular trajectory but also to general free-form trajectory. The presented formula is derived for both equally angular ray and equally spaced collinear detectors which are currently used in practice. Furthermore, a new plus-minus weighting scheme was proposed to handle the redundant projection data problem.

This paper is organized as follows. In section 2, we describe the derivation of the general fan-beam image reconstruction formula. The presented plus-minus weighting scheme that handle the redundant projection data problem with free-form source trajectory is described in section 3. Reconstruction results obtained from computer-simulated data of the Shepp-Logan head phantom are presented in section 4. Discussions and conclusions are presented in section 5.

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2. THEORY

This section discusses the derivation of the general fan-beam reconstruction formulas which starts from the parallel-beam reconstruction algorithm.

Reconstruction from parallel-beam projections

The 2D object density to be reconstructed is denoted as $f(\vec{x})$, the parallel-beam projections are given by

$$P(l,\vec{\theta}) = \int_R d\vec{x} f(\vec{x})\delta(\vec{x}\cdot\vec{\theta} - l) \tag{1}$$

where $l \in R$, $\vec{\theta}$ is a unit vector and δ is the Dirac delta function. The $P(l,\vec{\theta})$ is also known as the Radon transform of $f(\vec{x})$. The classical Filtered Back-Projection (FBP) reconstruction of $f(\vec{x})$ from its Radon transform obtained through two steps in [7].

$$f(\vec{x}) = \int_0^{\pi} d\phi P_F(l,\vec{\theta})\Big|_{l=\vec{\theta}\cdot\vec{x}} \tag{2}$$

where $\vec{\theta}\cdot\vec{x}$ is the scalar product of vectors $\vec{x}$ and $\vec{\theta}$, and $\vec{\theta} = (-\sin\phi, \cos\phi)$.

The filter to be applied to the projections $P(l,\vec{\theta})$ to obtain $P_F(l,\vec{\theta})$ is the 1D ramp filter:

$$h_F(l) = \int_{-\infty}^{+\infty} d\sigma |\sigma| e^{i2\pi\sigma l} \tag{3}$$

This filter is applied to equation (l):

$$P_F(l,\vec{\theta}) = \int dl' h_F(l-l') P(l',\vec{\theta}) \tag{4}$$

The implementation of the Ramp filter is equivalent to the successive implementation of a Hilbert transform followed by a derivation with $|\sigma| = (i2\pi\sigma)(-i\,\mathrm{sgn}(\sigma))/2\pi$.

$$P_F(l,\vec{\theta}) = \frac{1}{2\pi}\frac{\partial}{\partial l} P_H(l,\vec{\theta}) \tag{5}$$

Where $P_H(l,\vec{\theta})$ is the Hilbert transform of $P(l,\vec{\theta})$, and $\mathrm{sgn}(\cdot)$ is a sign function at fixed direction $\vec{\theta}$. Namely,

$$P_H(l,\vec{\theta}) = \int_R dl' h_H(l-l') P(l',\vec{\theta}) \tag{6}$$

with

$$h_H(l) = -\int_{-\infty}^{+\infty} d\sigma i\,\mathrm{sgn}(\sigma) e^{i2\pi\sigma l} \tag{7}$$

Using the above definitions, the FBP formula in equation (2) can be rewritten in the following form

$$f(\vec{x}) = \frac{1}{2\pi}\int_0^{\pi} d\phi \frac{\partial}{\partial l} P_H(l,\vec{\theta})\Big|_{l=\vec{\theta}\cdot\vec{x}} \tag{8}$$

The equations (2) and (8) that present the reconstruction of $f(\vec{x})$ from parallel-beam projections will be extensively used through this paper.

Reconstruction from fan-beam projections

The reconstruction from fan-beam projections is investigated for free-form trajectory consisting of finitely many convex and smooth curves (continuously differentiable). The trajectory is parameterized by a scalar λ varying in some intervals or union of disjoint intervals Λ of R, and the source point is given by $\vec{a}(\lambda)$. The vector tangent to the trajectory at $\vec{a}(\lambda)$ is $\vec{a}'(\lambda) = \partial\vec{a}(\lambda)/\partial\lambda$. Because the trajectory is convex and smooth, $\vec{a}'(\lambda)$ exists and is non-zero. The fan-beam projection is

$$g(\lambda,\vec{\alpha}) = \int_0^{+\infty} dt f(\vec{a}(\lambda) + t\vec{\alpha}) \tag{9}$$

which is obtained by varying $\vec{\alpha}$ in the set of unit vectors in the XY plane. Equation (9) does not present any assumptions regarding the detector geometry. Hamaker *et.al.* in [6] presents the relations between the fan-beam x-ray projections and the Radon transform, a method to access the Hilbert transform of the 2D Radon transform $P_H(l,\vec{\theta})$ from the Hilbert transform of the fan-beam projections $g_H(\lambda,\vec{\beta})$. The value of $P_H(l,\vec{\theta})$ on any line l passing through an X-Ray source point $\vec{a}(\lambda)$ can be obtained from the fan-beam projections $g_H(\lambda,\vec{\beta})$ by:

$$P_H(l,\vec{\theta})\Big|_{l=\vec{a}(\lambda)\cdot\vec{\beta}} = g_H(\lambda,\vec{\beta}) \tag{10}$$

where

$$g_H(\lambda,\vec{\beta}) = -\int d\vec{\alpha} h_H(\vec{\alpha}\cdot\vec{\beta}) g(\lambda,\vec{\alpha}) \tag{11}$$

with line l which is orthogonal to the unit vector $\vec{\beta}$ and contains $\vec{a}(\lambda)$, satisfies $l = \vec{a}(\lambda)\cdot\vec{\beta}$. We derivate both sides of the equation (10) with respect to λ,

$$\frac{\partial}{\partial\lambda} g_H(\lambda,\vec{\beta}) = \frac{\partial}{\partial\lambda}(P_H(\vec{a}(\lambda)\cdot\vec{\beta},\vec{\theta})) = (\vec{a}'(\lambda)\cdot\vec{\beta})\frac{\partial}{\partial l} P_H(l,\vec{\theta})\Big|_{l=\vec{a}(\lambda)\cdot\vec{\beta}} \tag{12}$$

then by using equation (12) we rewrite equation (8) to get a rebinning FBP reconstruction algorithm. The reconstruction of $f(\vec{x})$ is from the fan-beam projection $g(\lambda,\vec{\alpha})$ by the following sequence $g(\lambda,\vec{\alpha}) \to g_H(\lambda,\vec{\beta}) \to P_H(l,\vec{\theta}) \to Equation\,(8) \to f(\vec{x})$. Theoretically this algorithm is not only used to exactly reconstruct the whole object, but also used to exactly reconstruct the region of interest (ROI) if and only if all lines l passing through the ROI intersect the source trajectory $\vec{a}(\lambda)$. Note that we do not restrict the algorithm to any special source trajectory or any special detector geometry. Mathematically, the general reconstruction formula is given as follow

$$P^1(\lambda,\vec{\beta}) = \frac{1}{\vec{a}'(\lambda)\cdot\vec{\beta}}\frac{\partial}{\partial\lambda}\int d\vec{\alpha} h_H(\vec{\beta}\cdot\vec{\alpha}) g(\lambda,\vec{\alpha}) \tag{13-1}$$

$$P^2(\lambda,\vec{\beta}) = W(\lambda,\vec{\beta}) P^1(\lambda,\vec{\beta}) \tag{13-2}$$

$$f(\vec{x}) = \frac{1}{4\pi}\int_\wedge\int \frac{1}{\|\vec{x}-\vec{a}(\lambda)\|}\delta(\vec{\alpha}\cdot\vec{\beta})\left|\vec{a}'(\lambda)\cdot\vec{\beta}\right| P^2(\lambda,\vec{\beta}) d\vec{\beta} d\lambda \tag{13-3}$$

while $\vec{\alpha}$ is the unit vector defined as

$$\vec{\alpha} = \frac{\vec{x}-\vec{a}(\lambda)}{|\vec{x}-\vec{a}(\lambda)|} \tag{14}$$

The weighting function $W(\lambda,\vec{\beta})$ normalizing the redundancy will be introduced later. In fact, the $P^1(\lambda,\vec{\beta})$ is equivalent to the $P_F(l,\vec{\theta})$. The back-projection (13-3) is from the Ramp filtered fan-beam projection (13-1) which is obtained by once Hilbert transform implementation and once derivative implementation on the projection data. The reconstruction algorithm is equivalent to the FBP-style algorithm proposed by Noo *et.al* in [2].

3. PLUS-MINUS WEIGHTING SCHEME

During projection data acquisition, if a single projection ray intersects the source trajectory more than once, the problem of data redundancy should be considered. In such case, it is required to make an explicit choice of the weighting function to dispose the redundant projection data. Normally the weight function $W(\lambda,\vec{\beta})$ accounts for this redundancy in [8-9].

$$\sum_{i=0}^{N} W(\lambda_i,\vec{\beta}) = 1 \tag{15}$$

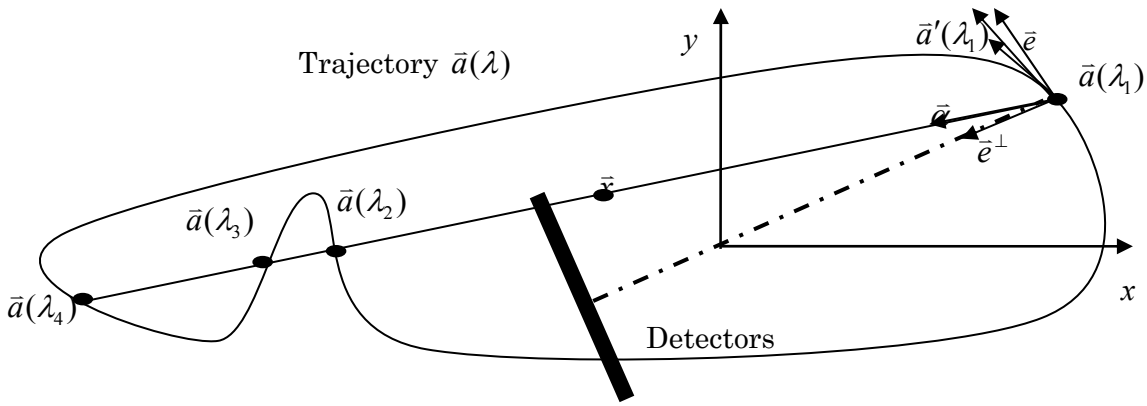


Figure 1. The geometry of a free-form source trajectory

Let N be the number of the intersections of a projection ray line with the whole source trajectory. For every line passing through the reconstruction object, $N \geq 1$. The intersections are labeled as $\vec{a}(\lambda_i), i = 1, \cdots, N$. There are several simple choices to satisfy the equation (15). For example, we can use $W(\lambda_i, \vec{\beta}) = 1/N$ or let one of the $W(\lambda_i, \vec{\beta})$ is 1 and the remaining are 0. However, these methods may not be the most appropriate ones when the source trajectory is free-form type. Because it is difficult to compute the number of intersection for each X-Ray line intersects the whole trajectory which is irregular or disjoint. We proposed a plus-minus weighting scheme by giving each intersection a sign as -1 or +1 instead of calculation the number of intersection. Finally compute the summation of all of successive signs to satisfy equation (15). When the source trajectory is closed, the number of intersection is even and it is required to multiply with a factor of 1/2 to satisfy equation (15). Correspondingly when the source trajectory is open, the number of intersection is odd. The weighting function for closed trajectory is given as follow,

$$W(\lambda_i, \vec{\beta}) = \frac{1}{2}\operatorname{sgn}(\vec{a}'(\lambda_i) \cdot \vec{\beta})\operatorname{sgn}(\vec{e} \cdot \vec{\beta})\operatorname{sgn}(\vec{\alpha} \cdot \vec{e}^{\perp}) \tag{16}$$

the weighting function for open trajectory is given as follow,

$$W(\lambda_i, \vec{\beta}) = \operatorname{sgn}(\vec{a}'(\lambda_i) \cdot \vec{\beta})\operatorname{sgn}(\vec{e} \cdot \vec{\beta})\operatorname{sgn}(\vec{\alpha} \cdot \vec{e}^{\perp}) \tag{17}$$

while vectors $\vec{e}$ and $\vec{e}^{\perp}$ is a pair of perpendicular unit vectors. The direction of $\vec{e}^{\perp}$ is from the intersection to the coordinate origin point, the direction of $\vec{e}$ is a clockwise 90° rotation of $\vec{e}^{\perp}$. The $\operatorname{sgn}(\cdot)$ is a sign function with value +1 and -1 when the angle of two vectors is less than or greater than 90° respectively. In fact the result of $W(\lambda_i, \vec{\beta})$ is also only just +1 and -1, one by another. For the first item of the weighting function $\operatorname{sgn}(\vec{a}'(\lambda_i) \cdot \vec{\beta})$, its value is +1 and -1 one by one, as each pair of two near intersections the directions of $\vec{a}'(\lambda)$ would be more than 90°. For all the intersection points beside on one half of the object, the result value of the second item $\operatorname{sgn}(\vec{e} \cdot \vec{\beta})$ and third item $\operatorname{sgn}(\vec{\alpha} \cdot \vec{e}^{\perp})$ in weighting function would be the same as +1 or -1. For the rest intersections on the other half side of the object, the corresponding value is always opposite value as -1 or +1. For each intersection the multiplication of the three items in equation (16) and equation (17) is always +1 or -1, finally summation all of the succession intersections would be 1, which means equation (16) and equation (17) satisfies the weighting redundant condition of equation (15). For a special source trajectory shown in figure 1, the projection ray l intersects the trajectory four times. The corresponding weighting function is given in tablet 1. As the trajectory is closed, by using the equation (16) we can get the result of weighting function $\sum_{i=1}^{4} W(\lambda_i, \vec{\beta}) = 1$.

When we eliminate the left side part of the closed trajectory, it became to an open trajectory. The same projection ray l will only have three intersections with the open trajectory, there is no intersection $\vec{a}(\lambda_4)$. The corresponding application of weighting function is given in tablet 2. We can get the result of weighting function $\sum_{i=1}^{3} W(\lambda_i, \vec{\beta}) = 1$ by using the equation (17).

Table 1 Application example of weighting function (16).

	$\vec{a}(\lambda_1)$	$\vec{a}(\lambda_2)$	$\vec{a}(\lambda_3)$	$\vec{a}(\lambda_4)$
$\mathrm{sgn}(\vec{a}'(\lambda)\cdot\vec{\beta})$	+1	+1	-1	+1
$\mathrm{sgn}(\vec{e}\cdot\vec{\beta})$	+1	+1	+1	+1
$\mathrm{sgn}(\vec{\alpha}\cdot\vec{e}^{\perp})$	+1	+1	+1	+1
$W(\lambda_i, \vec{\beta})$	+0.5	+0.5	-0.5	+0.5

Table 2 Application example of weighting function (17).

	$\vec{a}(\lambda_1)$	$\vec{a}(\lambda_2)$	$\vec{a}(\lambda_3)$
$\mathrm{sgn}(\vec{a}'(\lambda)\cdot\vec{\beta})$	+1	+1	-1
$\mathrm{sgn}(\vec{e}\cdot\vec{\beta})$	+1	+1	+1
$\mathrm{sgn}(\vec{\alpha}\cdot\vec{e}^{\perp})$	+1	+1	+1
$W(\lambda_i, \vec{\beta})$	+1	+1	-1

4. NUMERICAL EXPERIMENTS

In this section a simulation study was implemented to evaluate the general FBP fan-beam reconstruction formula for free-form trajectory using the 2D Shepp-Logan phantom with image matrix size of 512x512 pixels.

Source trajectory and implementation details

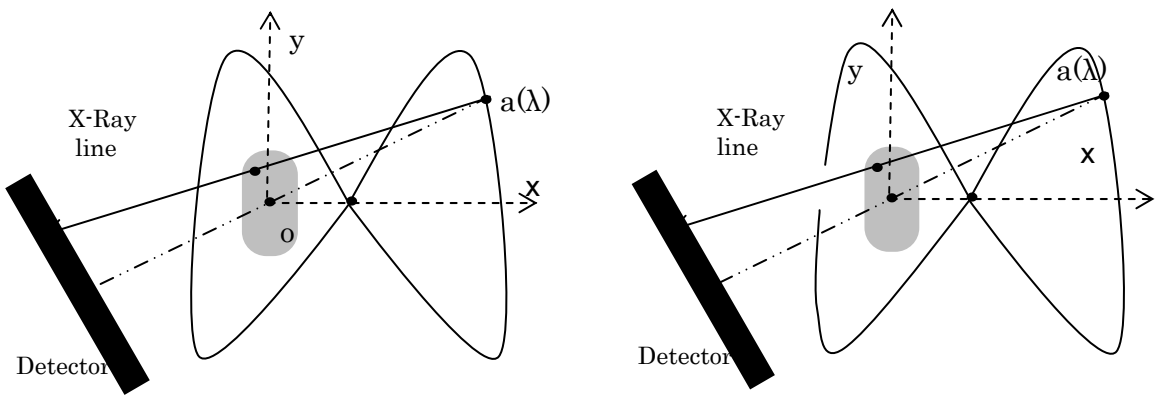
The equally spaced collinear rays detectors and butterfly source trajectory were selected as the simulation detectors and a sample of free-form trajectory, respectively. The detectors were assumed to be always orthogonal to the source-to-original line. We horizontally shift the cross point of the trajectory by half of the radius R for the calculation convenience, so the closed trajectory is defined as $\vec{a}(\lambda) = (R\sin(\lambda) + R/2, R\sin(2\lambda)) : \lambda \in (0^\circ, 360^\circ)$. The geometry is shown in figure 3(a). We select the R as 600*mm*, and the field of view was a circle of radius 256*mm*. The projection data were computed for 1024 bins and 1024 angles, and the detectors were consisted of 1024 channels with 1*mm* channel size. As the weighting function can be used to both closed and open trajectory, we also set the open butterfly trajectory as $\vec{a}(\lambda) = (R\sin(\lambda) + R/2, R\sin(2\lambda)) : \lambda \in (0^\circ, 260^\circ) \bigcup (280^\circ, 360^\circ)$. The geometry is shown in figure 3(b). We used the weighting function in equation (16) and (17) for closed and open butterfly trajectories respectively.

In the general reconstruction formula there is a derivative calculation on the projection data and the distance of source-to-detectors. A simple three-point estimation is a possible way to compute the numerical differentiation through $(x-h, f(x-h))$ and $(x+h, f(x+h))$, h is the unit interval of x. The derivative of $f(x)$ at x is

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} \tag{18}$$

Results

From the fan-beam projection data, we used our general reconstruction formula to reconstruct images, which are shown in figure 5(c) and (e). The reconstructed images are identical to the original phantom that is shown in figure 5(a). These

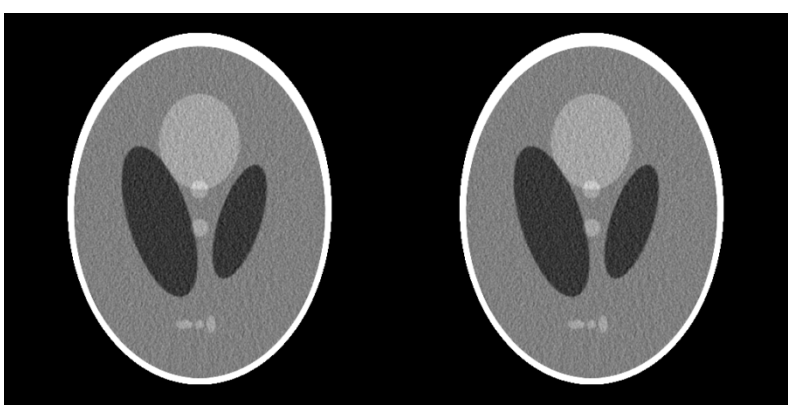


(a) (b)

Figure.3 Butterfly trajectory geometry. (a): closed, (b): open

results verify our theoretical prediction for free-form trajectory that the proposed plus-minus weighting function can handle the redundant projection data problem. Reconstruction results are shown in Fig.4 and Fig.5 in noised and noise-free case respectively. From these results we can see the correctly reconstructed images by using the open and closed butterfly trajectory both in noised and noised-free projection data.

However, we also found that there is a relatively small artifact in the reconstructed image by using the butterfly trajectory appears at profiles (Figures 4(b), (d) and (e)). The artifact is most possibly happened during calculation of plus-minus weighting function. Each item of the weighting function is calculated on the corresponding position of source point and the object point. We want to get the exact position of intersection points produced by the projection ray intersecting with the source trajectory. Unfortunately, in practice computer simulation, it is difficult to get the exact position because of the discretization errors. In another word, the corresponding intersections may be not on a projection ray, then it will affect the sign operator of the weighting function. But when we increase the angle sample the discretization errors can be sharply reduced.



(a) (b)

Figure 4. Reconstructed image from Poisson noisy projection data with photon $N = 6.5 \times 10^{11}$. (a) Reconstructed image from closed butterfly trajectory. (b) Reconstructed image from open butterfly trajectory

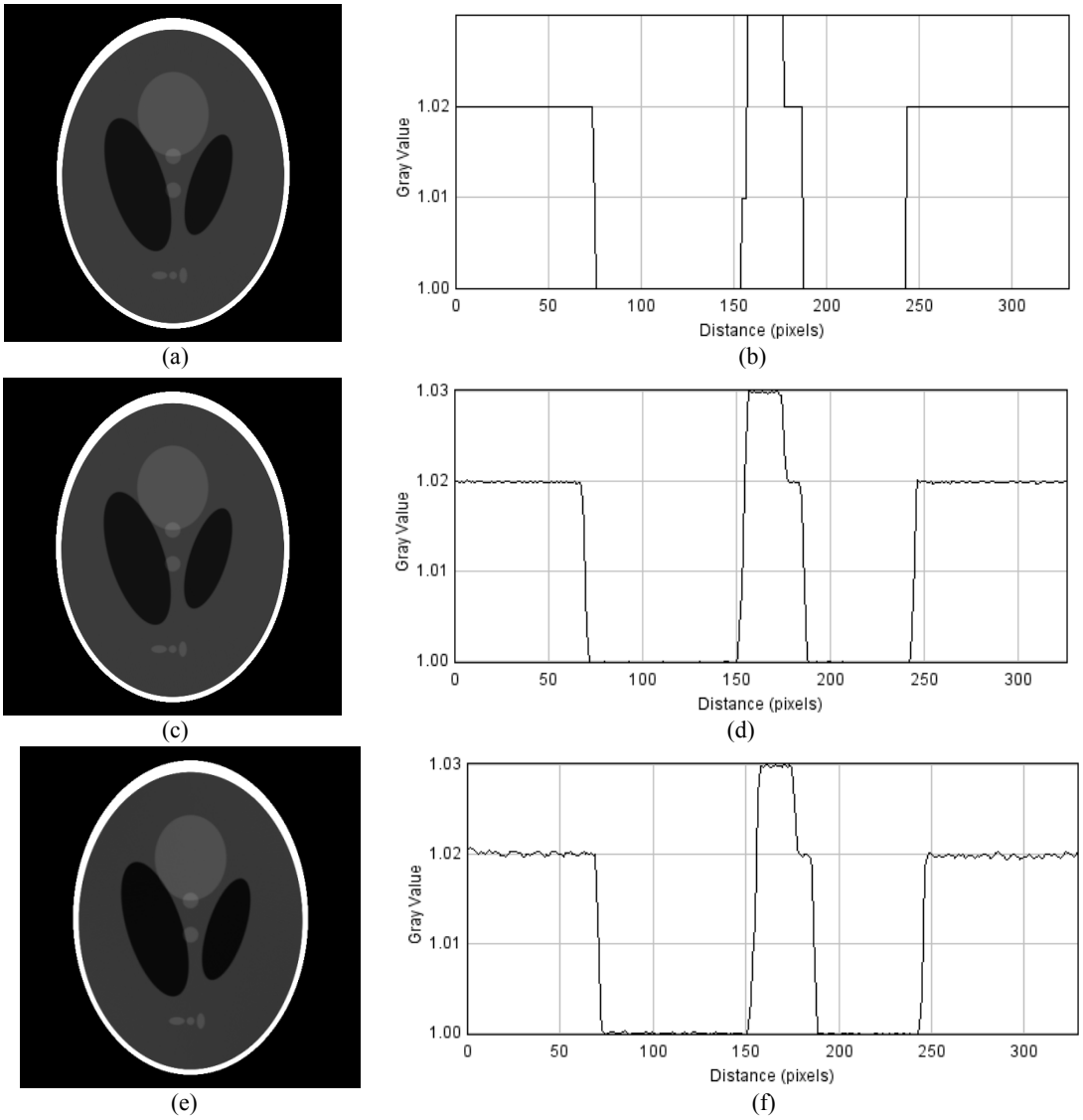


Figure. 5 Reconstruct images and the same line profile.

(a) Shepp-Logan Phantom, (b) Profile of line l in (a), (c) Reconstructed image from closed butterfly trajectory, (d) Profile of line L in (c), (e) Reconstructed image from open butterfly trajectory , (f) Profile of line l in (e). l line: $y = 280, x \in [90,420]$. The display scale is $[0.95,1.10]$.

5. CONCLUSIONS

In this paper we rewrote the classical fan-beam reconstruction formula in 2DCT from circular source trajectory, given an exact and efficient general fan-beam reconstruction algorithm from free-form source trajectory. In order to apply the proposed algorithm to free-form source trajectory which is not only suitable for closed but also for open trajectory, we proposed a new plus-minus weighting scheme to handle the redundant projection data problem. The new weighting scheme let us do not need to calculate the number of intersections of each projection ray intersect the source trajectory. With the proposed plus-minus weighting function there is a freedom to choose any source trajectory, which can be appropriate to specific objects in different imaging applications. One possible case is the non-destructive testing in industry, because the testing object is not always the same. Even in the medical imaging application while the cardiac is

beating during projection data acquisition, the circular trajectory is needed to change to free-form trajectory by rigid transform before reconstruction.

Furthermore, we gave the detail formula for equally angular and equally space detectors. The simulation data demonstrated that fan-beam reconstruction is highly accurate, with acceptable artifacts of discretization. It is also possible to use non-smooth trajectory with sharp corner, so without any source trajectory limitation, we confirmed that the proposed algorithm would be useful in medical and industry applications to find a more appropriate source trajectory. The practical implications of our results could be significant for all applications of CT where 2D reconstruction is still used.

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